Mathematical Tools

Forces (and Fields) are vector quantities..

We need to know how to handle them

- **Vector algebra**
- **Vector calculus**

We need to develop some mathematical formulations to work with…

… specially some general tools of vector calculus!

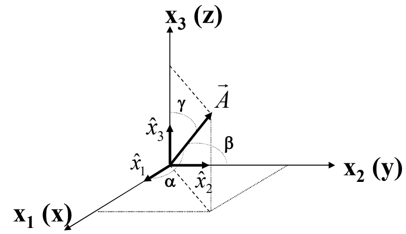
- **Gauss' theorem**
- **Stoke's theorem**

1

Mathematical Tools

I assume that you know..

- **what a vector is?**



$$\vec{A} = A_{x_1}\hat{x}_1 + A_{x_2}\hat{x}_2 + A_{x_3}\hat{x}_3$$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$\vec{A} : (A_x, A_y, A_z)$

$(A\cos\alpha, A\cos\beta, A\cos\gamma)$

Magnitude of $\vec{A}$

$$|\vec{A}| = \sqrt{A_x^{\,2} + A_y^{\,2} + A_z^{\,2}}$$

2

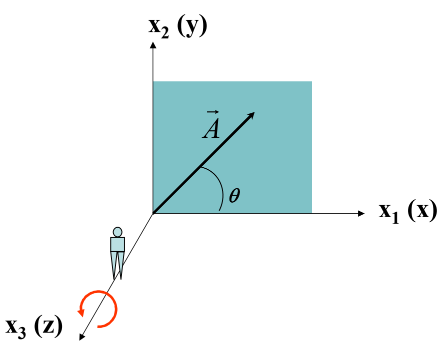
Mathematical Tools

I assume that you know..

- **what a vector is?**

$\vec{A} : (A_x, A_y)$

$(A\cos\theta, A\sin\theta)$



3

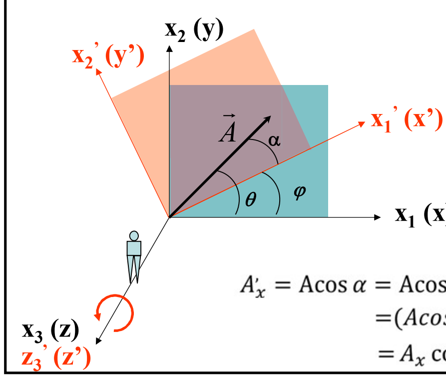
Mathematical Tools

I assume that you know..

- **what a vector is?**

$\vec{A} : (A_x, A_y)$

$(A\cos\theta, A\sin\theta)$



$x_1 \rightarrow x_1'$

$x_2 \rightarrow x_2'$

?

$\vec{A} : (A'_x, A'_y)$

$(A\cos\alpha, A\sin\alpha)$

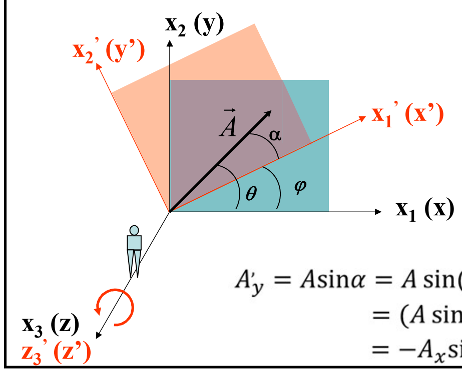
$$A'_x = A\cos\alpha = A\cos(\theta - \varphi)$$
$$= (A\cos\theta)\cos\varphi + (A\sin\theta)A\sin\varphi$$
$$= A_x\cos\varphi + A_y A\sin\varphi$$

4

Mathematical Tools

I assume that you know..

- **what a vector is?**



$\vec{A}:(A_x, A_y)$
(A cos θ, A sin θ)

$x_1 \rightarrow x_1'$

$x_2 \rightarrow x_2'$

?

$\vec{A}:(A'_x, A'_y)$
(A cos α, A sin α)

$$\begin{aligned} A'_y &= A\sin\alpha = A\sin(\theta-\varphi) \\ &= (A\sin\theta)\cos\varphi - \sin\varphi(A\cos\theta) \\ &= -A_x\sin\varphi + A_y\cos\varphi \end{aligned}$$

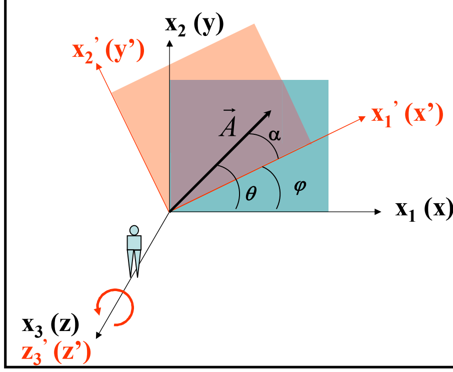
5

Mathematical Tools

I assume that you know..

- **what a vector is?**

$$A'_x = A_x\cos\varphi + A_y\sin\varphi$$
$$A'_y = -A_x\sin\varphi + A_y\cos\varphi$$



$\vec{A}:(A_x', A_y')$

$\vec{A}:(A_x, A_y)$

$$\begin{pmatrix} A_x' \\ A_y' \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$\begin{pmatrix} A_x' \\ A_y' \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$a_{12} = \cos(x_1', x_2) = \cos(\frac{\pi}{2} - \varphi)$$

$$a_{21} = \cos(x_2', x_1) = \cos(\varphi + \frac{\pi}{2})$$

6

Mathematical Tools

I assume that you know..

- **what a vector is?**

Must have a unique magnitude

Must have a unique direction

Components must follow the transformation relation

7

Mathematical Tools

I assume that you know..

- **what a vector is?**
- **vector algebra**

  dot (scalar) product

  cross (vector) product

  scalar triple product

  $$\vec{A}.(\vec{B}\times\vec{C}) = \vec{B}.(\vec{C}\times\vec{A}) = \vec{C}.(\vec{A}\times\vec{B})$$

  vector triple product

  $$\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}.\vec{C}) - \vec{C}(\vec{A}.\vec{B})$$

- **line, surface & volume integration**

8

Mathematical Tools

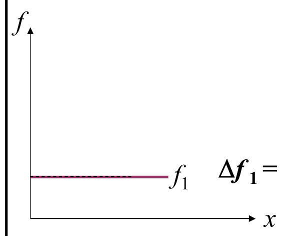
I shall start with vector calculus…

…first, vector differentiation!

Function of One Variable: Ordinary Derivative

Consider a function f, which is a linear function of a single variable x.

Question: What would be change in f for a given change in the variable x ? $\Delta f = f(x + \Delta x) - f(x)$



$\Delta f_1 = 0$ $\quad df_1 = 0$

Function of One Variable: Ordinary Derivative

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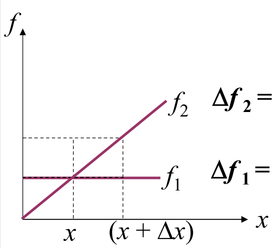


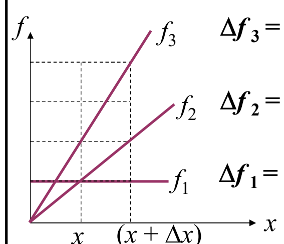
$\Delta f_2 = \Delta x$ $\quad df_2 = dx$

$\Delta f_1 = 0$ $\quad df_1 = 0$

Function of One Variable: Ordinary Derivative

Consider a function f, which is a linear function of a single variable x.

Question: What would be change in f for a given change in the variable x ? $\Delta f = f(x + \Delta x) - f(x)$



$\Delta f_3 = 2\Delta x$ $\quad df_3 = 2\,dx$ $\quad y = f_3(x) = 2x$ $\quad \dfrac{dy}{dx} = 2$

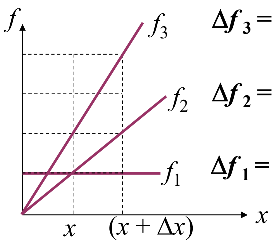
$\Delta f_2 = \Delta x$ $\quad df_2 = dx$ $\quad y = f_2(x) = x$ $\quad \dfrac{dy}{dx} = 1$

$\Delta f_1 = 0$ $\quad df_1 = 0$ $\quad y = f_1(x) = C$ $\quad \dfrac{dy}{dx} = 0$

Function of One Variable: Ordinary Derivative

Consider a function ***f***, which is a linear function of a single variable ***x***.

Question: What would be change in ***f*** for a given change in the variable ***x*** ? $\Delta f = f(x + \Delta x) - f(x)$



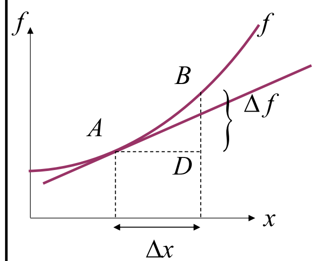
$\Delta f_3 = 2\Delta x$	$df_3 = 2\,dx$	$y = f_3(x) = 2x$	$\frac{dy}{dx} = 2$
$\Delta f_2 = \Delta x$	$df_2 = 1\,dx$	$y = f_2(x) = x$	$\frac{dy}{dx} = 1$
$\Delta f_1 = 0$	$df_1 = 0\,dx$	$y = f_1(x) = C$	$\frac{dy}{dx} = 0$

$$df = \left(\frac{df}{dx}\right) dx$$

Function of One Variable: Ordinary Derivative

Consider a function ***f***, which is a non-linear function of a single variable ***x***.

Question: What would be change in ***f*** for a given change in the variable ***x*** ? $\Delta f = f(x + \Delta x) - f(x)$



$$df = \left(\frac{df}{dx}\right) dx$$

Is this definition universal?

Yes, if $\Delta x \to 0$

Function of Many Variable: Partial Derivative

Consider a function ***f***, which is now a function of three variables ***x, y, z***.

Example: ***T(x, y, z)*****:** Essentially a scalar point function

Question: How fast ***T*** varies as we move from one point to another?

Note: derivative gave information about how fast the function changes if we move a little distance**…..**

….. but the situation here is complicated

change in ***T*** depends which direction the second point is from the starting point (i.e., which direction we are moving)!

going straight up → ***T*** may increase fairly rapidly
going horizontal → ***T*** may not change at all
going in any other direction → rate of change would be intermediate

Function of Many Variable: Partial Derivative

Consider a function ***f***, which is now a function of three variables ***x, y, z***.

Example: ***T(x, y, z)*****:** Essentially a scalar point function

Question: How fast ***T*** varies as we move from one point to another?

Answer: it depends on which direction we are moving

….. It has infinite many answers, one for each direction

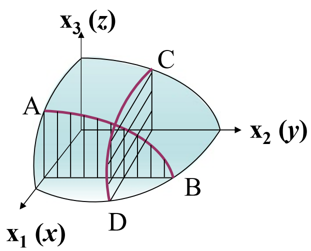
It looks like, finding change in $T(x, y, z)$ is too complicated?

It is not

We need to know function of multi-variables more deeply and how to handle them?

Function of Many Variable: Partial Derivative

Consider a simple case, $z = f(x, y)$ essentially a two dimensional case.



$$\frac{\partial z}{\partial y} = \lim_{\Delta y \to 0} \frac{z(x, y+\Delta y) - z(x, y)}{\Delta y}$$

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \to 0} \frac{z(x+\Delta x, y) - z(x, y)}{\Delta x}$$

Function of Many Variable: Partial Derivative

Example of $z = f(x, y)$: $f(x,y) = 2x^3y^2 + y^3$

$$\frac{\partial f}{\partial x} = ?$$

$$\frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial f}{\partial x} = ?$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\frac{\partial f}{\partial x} = ?$$

Function of Many Variable: Partial Derivative

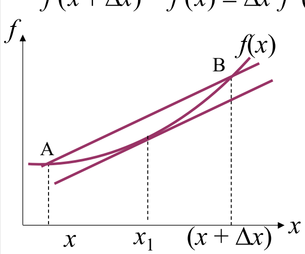
Come back to the original question: change in $f(x, y, z)$?

$$\Delta f(x,y) = f(x+\Delta x, y+\Delta y) - f(x,y)$$

$$= \underline{f(x+\Delta x, y+\Delta y) - f(x+\Delta x, y)} + \underline{f(x+\Delta x, y) - f(x,y)}$$

Using the Mean Value Theorem of calculus

$$f(x+\Delta x) - f(x) = \Delta x\, f'(x_1)$$



$$\Delta f(x,y) = \frac{\partial f(x_1, y)}{\partial x} dx + \frac{\partial f(x+\Delta x, y_1)}{\partial y} dy$$

$$for\ \Delta x, \Delta y \to 0$$

$$\Delta f(x,y) = \frac{\partial f(x,y)}{\partial x} dx + \frac{\partial f(x,y)}{\partial y} dy$$

Function of Many Variable: Partial Derivative

$$\Delta f(x,y) = \frac{\partial f(x,y)}{\partial x} dx + \frac{\partial f(x,y)}{\partial y} dy$$

<u>Extending to three dimension</u>

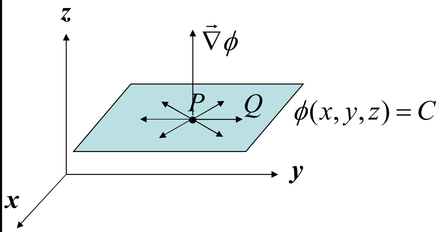
$$\boxed{\begin{aligned} df(x,y,z) &= \frac{\partial f(x,y,z)}{\partial x} dx + \frac{\partial f(x,y,z)}{\partial y} dy + \frac{\partial f(x,y,z)}{\partial z} dz \\ df(x,y,z) &= \left(\frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z}\right).\left(dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}\right) \\ &= \left(\vec{\nabla} f\right)\left(d\vec{r}\right) \end{aligned}}$$

$\vec{\nabla} f$ **is defined as Gradient of a scalar function *f*.**

$$\vec{\nabla} \equiv \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \qquad \vec{\nabla} f \neq f\,\vec{\nabla}$$

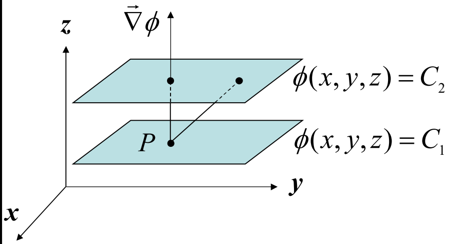
$\vec{\nabla}$ **(scalar field)** $\longrightarrow$ **(vector field)**

Physical Significance of Gradient



$$d\phi = \vec\nabla\phi\cdot d\vec r$$
$$= 0$$
$$\boxed{\vec\nabla\phi \perp d\vec r}$$

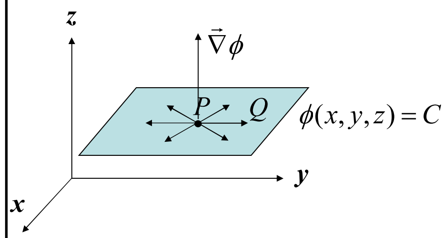
Universal Fact



$$d\phi = C_2 - C_1 = \vec\nabla\phi\cdot d\vec r$$

For a given $d\phi$, $|d\vec r|$ is min in the direction of $\vec\nabla\phi$

Physical Significance of Gradient



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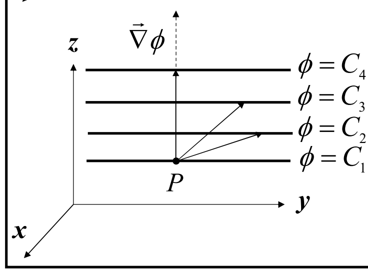
Universal Fact



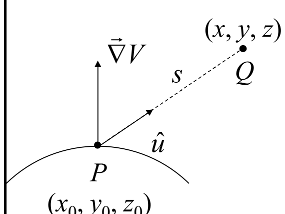
Let's fix, $|d\vec r|$

$\vec\nabla\phi$ indicates the direction of maximum change in ϕ

Physical Significance of Gradient

Consider the gravitational field as the scalar field represented by gravitational potential energy (V). $V = f(x,y,z) = f(s)$

Question: if we move in some other direction, how should we find *dV*?



$$\hat u = a\hat x + b\hat y + c\hat z$$
$$P\vec Q = s\hat u$$
$$= as\hat x + bs\hat y + cs\hat z$$
$$= (x-x_0)\hat x + (y-y_0)\hat y + (z-z_0)\hat z$$

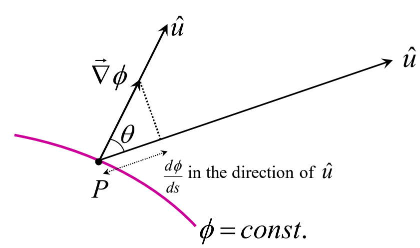
$$\left.\begin{aligned} x &= x_0 + as \\ y &= y_0 + bs \\ z &= z_0 + cs \end{aligned}\right\}$$ **Parametric equation of the line PQ**

$$\frac{dV}{ds} = \frac{\partial V}{\partial x}\frac{dx}{ds} + \frac{\partial V}{\partial y}\frac{dy}{ds} + \frac{\partial V}{\partial z}\frac{dz}{ds} = \frac{\partial V}{\partial x}a + \frac{\partial V}{\partial y}b + \frac{\partial V}{\partial z}c = \vec\nabla V\cdot\hat u$$

Directional derivative $\frac{dV}{ds} = \vec\nabla V\cdot\hat u$

Physical Significance of Gradient

Directional derivative $\frac{d\phi}{ds} = \vec\nabla\phi\cdot\hat u = |\vec\nabla\phi|\cos\theta$



$$\left.\frac{d\phi}{ds}\right|_{max} = |\vec\nabla\phi| \quad \text{when } \theta = 0$$

Physical Significance of Gradient

Example: Find the directional derivative of $f(x,y) = x^2 y + xz$ at (1, 2, -1) in the direction of $\vec{A} = 2\hat{x} - 2\hat{y} + \hat{z}$.

$$\boxed{\vec{\nabla} f \cdot \hat{u}\Big|_{(1,2,-1)}}$$

$$\vec{\nabla} f = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \right)\left(x^2 y + xz\right)$$

$$= \hat{x}(2xy+z) + x^2\hat{y} + x\hat{z}$$

$$\hat{u} = \frac{2\hat{x} - 2\hat{y} + \hat{z}}{\sqrt{4+4+1}} = \frac{1}{3}\left(2\hat{x} - 2\hat{y} + \hat{z}\right)$$

$$\vec{\nabla} f \cdot \hat{u}\Big|_{(1,2,-1)} = \frac{2}{3}(2xy+z) - \frac{2}{3}x^2 + \frac{1}{3}x\Big|_{(1,2,-1)} = \frac{5}{3}$$ ◀

Physical Significance of Gradient

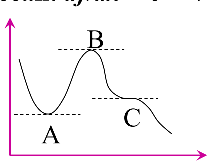
Example: Find $\vec{\nabla} f$ where $f(x,y,z) = r = \sqrt{x^2+y^2+z^2}$.

Answer: $\vec{\nabla} r = \hat{r}$ ◀

Physical Significance of Gradient

What if Gradient vanishes? $\vec{\nabla}\phi = 0$

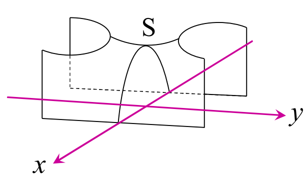
Recall: $df/dx = 0$ $\rightarrow$ stationary points (A, B, C)



A $\rightarrow$ local minima ($f'' > 0$)
B $\rightarrow$ local maxima ($f'' < 0$)
C $\rightarrow$ point of inflection ($f'' = 0$)

$\vec{\nabla}\phi = 0$ $\rightarrow$ **stationary points**

local minima
local maxima
saddle point (S)



The del operator

$$\vec{\nabla} \equiv \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

is an operator which "<u>acts upon</u>" a given function.

Possible ways it can act upon:
on a scalar function φ : $\vec{\nabla}\varphi$ (Gradient)
on a vector function $\vec{V}$: $\vec{\nabla}\cdot\vec{V}$ (Divergence)
$\vec{\nabla}\times\vec{V}$ (Curl)

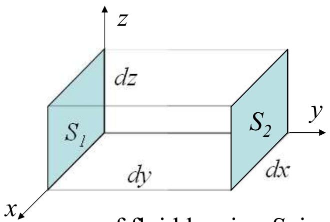
<u>Divergence</u>

$$\vec{\nabla}\cdot\vec{V} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \right)\cdot\left(\hat{x}V_x + \hat{y}V_y + \hat{z}V_z\right)$$

$$= \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Divergence: Physical Significance

Consider a region in which a compressible fluid $\rho(x,y,z)$ is flowing with a velocity $\vec{v} = v_x(x,y,z)\hat{x} + v_y(x,y,z)\hat{y} + v_z(x,y,z)\hat{z}$



mass of fluid entering through S_1 in a time interval dt = density × volume

$$= (\rho)\big|_{y=0}\, dxdz(v_y)\big|_{y=0}\, dt$$

$$= (\rho v_y)\big|_{y=0}\, dxdzdt$$

mass of fluid leaving S_2 in a time interval $dt = (\rho)\big|_{y=dy}\, dxdz(v_y)\big|_{y=dy}\, dt$

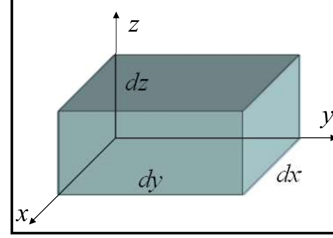
$$= (\rho v_y)\big|_{y=dy}\, dxdzdt$$

Net mass of fluid leaving the volume element in a time interval dt along y-direction: $= [(\rho v_y)\big|_{y=dy} - (\rho v_y)\big|_{y=0}]\, dxdzdt$

Divergence: Physical Significance

Net mass of fluid leaving the volume element in a time interval dt along y-direction: $= [(\rho v_y)\big|_{y=dy} - (\rho v_y)\big|_{y=0}]\, dxdzdt \quad = \dfrac{\partial(\rho v_y)}{\partial y} dydxdzdt$

Net mass of fluid leaving the volume element in a time interval dt along x-direction: $= \dfrac{\partial(\rho v_x)}{\partial x} dxdzdydt$



Net mass of fluid leaving the volume element in a time interval dt along z-direction: $= \dfrac{\partial(\rho v_z)}{\partial z} dzdxdydt$

Net mass of fluid leaving the volume element in a time interval dt :

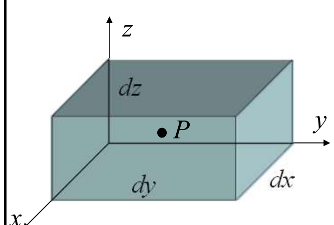
$$= \left(\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z}\right) dydxdzdt$$

Divergence: Physical Significance

Net flow out per unit time:

$$= \left(\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z}\right) dydxdz \quad = \vec{\nabla}\cdot(\rho\, \vec{v})\, dxdydz$$

Net flow out per unit time per unit volume: $= \vec{\nabla}\cdot(\rho\, \vec{v})$



Divergence is the measure of how much a vector field spreads out (diverges) from a given point

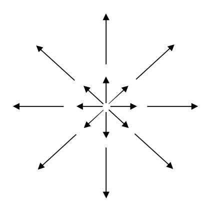
Divergence: Continued

$\vec{\nabla}\cdot(\vec{v})$ **= (net mass left – net mass entered)/volume/time**

Divergence measures flux of the vector field

$\vec{\nabla}\cdot(\vec{v}) = 0$ → **Vector field is solenoidal**

Examples



Large + ve divergence

$$\vec{V} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{\nabla}\cdot\vec{V} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 1 + 1 \;= 3$$

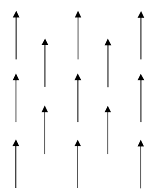
Divergence: Continued

$\vec{\nabla}\cdot(\vec{v})$ = (net mass left – net mass entered)/volume/time

Divergence measures flux of the vector field

$\vec{\nabla}\cdot(\vec{v}) = 0$ $\rightarrow$ **Vector field is solenoidal**

Examples



zero divergence

$$\vec{V} = c\hat{z}$$

$$\vec{\nabla}\cdot\vec{V} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(c)$$
$$= 0$$

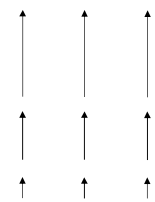
Divergence: Continued

$\vec{\nabla}\cdot(\vec{v})$ = (net mass left – net mass entered)/volume/time

Divergence measures flux of the vector field

$\vec{\nabla}\cdot(\vec{v}) = 0$ $\rightarrow$ **Vector field is solenoidal**

Examples



finite divergence

$$\vec{V} = z\hat{z}$$

$$\vec{\nabla}\cdot\vec{V} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z)$$
$$= 1$$

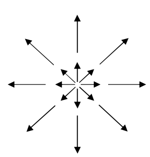
Divergence: Continued

$\vec{\nabla}\cdot(\vec{v})$ = (net mass left – net mass entered)/volume/time

Divergence measures flux of the vector field

$\vec{\nabla}\cdot(\vec{v}) = 0$ $\rightarrow$ **Vector field is solenoidal**

Examples $\vec{V} = \frac{\hat{r}}{r^2}$ **large divergence**



$$\vec{\nabla}\cdot\vec{V} = \vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\right) = \vec{\nabla}\cdot\left(\frac{\vec{r}}{r^3}\right) = \vec{\nabla}\cdot\left(\frac{x\hat{x}+y\hat{y}+z\hat{z}}{r^3}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right)$$

$$\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = \frac{\partial}{\partial x}\left[x\left(x^2+y^2+z^2\right)^{-3/2}\right] = \frac{1}{(x^2+y^2+z^2)^{3/2}} - \frac{3}{2}\frac{x(2x)}{(x^2+y^2+z^2)^{5/2}}$$

$$= \frac{1}{r^3} - \frac{3x^2}{r^5} + \frac{1}{r^3} - \frac{3y^2}{r^5} + \frac{1}{r^3} - \frac{3z^2}{r^5} = \frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5} = 0\,?$$

Curl

$$\vec{\nabla} \equiv \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

$$\vec{V} = \hat{x}V_x + \hat{y}V_y + \hat{z}V_z$$

$$\vec{\nabla}\times\vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_x & V_y & V_z \end{vmatrix}$$

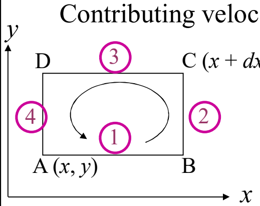
$$= \hat{x}\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right) + \hat{y}\left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right) + \hat{z}\left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)$$

$$\vec{\nabla}\times\vec{V} \neq \vec{V}\times\vec{\nabla}$$

Curl: Physical Significance

Consider the circulation of a fluid around a differential loop in the *xy*-plane.

Contributing velocity for this circulation: $\vec{v} = v_x(x,y)\hat{x} + v_y(x,y)\hat{y}$



Circulation along path "1" $= v_x(x,y)dx$

Circulation along path "2" $= v_y(x+dx,y)dy$

Circulation along path "3" $= -v_x(x,y+dy)dx$

Circulation along path "4" $= -v_y(x,y)dy$

Circulation along ABCDA

$$= v_x(x,y)dx + v_y(x+dx,y)dy - v_x(x,y+dy)dx - v_y(x,y)dy$$

37

Curl: Physical Significance

Circulation along ABCDA

$$= v_x(x,y)dx + v_y(x+dx,y)dy - v_x(x,y+dy)dx - v_y(x,y)dy$$

Using Taylor expansion $v_y(x+dx,y) = v_y(x,y) + \frac{\partial v_y}{\partial x}dx$

Circulation along ABCDA

$$= v_x\cancel{(x,y)}dx + v_y\cancel{(x,y)}dy + \frac{\partial v_y}{\partial x}dxdy - v_x\cancel{(x,y)}dx - \frac{\partial v_x}{\partial y}dxdy - v_y\cancel{(x,y)}dy$$

$$= \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)dxdy = (\vec{\nabla}\times\vec{V})_z\, dxdy$$

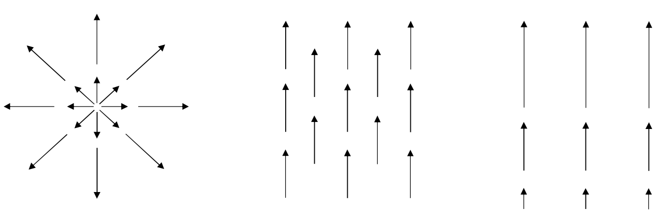
Curl of a vector filed is the measure of the circulation of the field in a particular plane

How much a field curl around the point in question

$\vec{\nabla}\times\vec{V} = 0 \;\rightarrow$ **Irrotational Field**

38

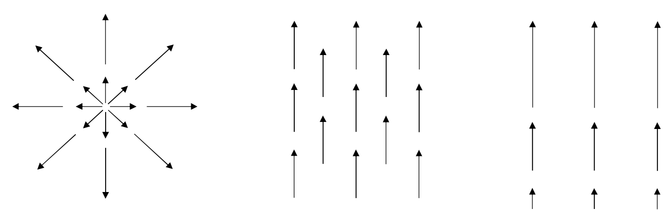
Curl: Example



$\vec{\nabla}\times\vec{V} = 0 \;\rightarrow$ **Irrotational Fields**

39

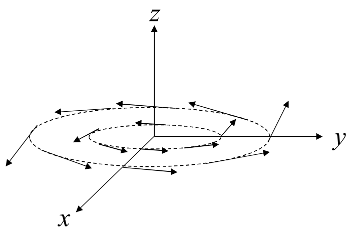
Curl: Example



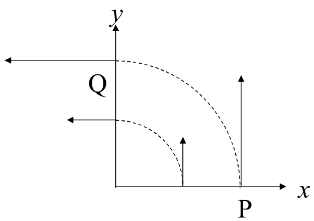
Home Work: Calculate curl for each field from first principle

40

Curl: Example



finite divergence



At P ($x = x, y = 0$): $x\hat{y}$

At Q ($x = 0, y = y$): $-y\hat{x}$

$$\vec{V} = -y\hat{x} + x\hat{y}$$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix}$$

$$= 2\hat{z}$$